COMP9444 Project Summary

<Mosquitos on HumanSkin Recognition>

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I. Introduction

The focus of this project is the application of deep-learning models in the automatic classification of mosquitoes that land or are swatted on humans. This is very important because mosquitoes remain one of the primary vectors of diseases including malaria, dengue fever, and Zika. This project performs through comparative study various deep-learning models, namely, MobileNet, ResNet50, VGG16, InceptionV3. and the categorical variables analyzed in terms of classification accuracy, training, and recognition speed, and used resources.

The main contributions of this study are:

1. Model Selection Guidance: A comparative analysis highlights MobileNet's suitability for resource-constrained, real-time applications, ResNet50 for precision-focused tasks, VGG16 for offline processing, and InceptionV3's limitations in this context.
2. Real-World Insights: Practical recommendations for selecting models based on specific application needs.
3. Performance Improvements: Strategies for enhancing models through data augmentation, optimization, and parameter tuning.

II. Related Work

Recent studies have increasingly applied deep learning, especially convolutional neural networks (CNNs), to image classification of mosquitoes. Pratik et al. (2019) used a multilayer CNN and ResNet50 to analyse the wing movement patterns of mosquitoes, achieving an accuracy of 80%-86%, focusing on dynamic behaviour rather than fixed states. Song-Quan et al. (2022) improved classification accuracy by performing transfer learning on a DSLR image dataset, but only achieved 77.75% accuracy due to limited diversity in the dataset.

Armin et al. (2021) used ResNet50 on community-collected data and obtained a 0.96 ROC AUC by using non-mosquito images as negative samples, but the study did not cover stationary behaviours such as injured or landing mosquitoes. Challenges remain in distinguishing morphologically similar species such as Aedes aegypti and Aedes albopictus, and in generalising model application. Increasing data volume and reference databases are needed to improve accuracy.

Despite the significant progress made in mosquito classification, there is currently no model that can solve the problem of mosquito classification in non-native habitats, nor is there a model that can use deep learning to accomplish such tasks, which highlights the need to develop reliable and deployable solutions.

III. Methods

Transfer Learning Approach
Advantages:

1. **Limited Data:** Utilizes pre-trained models to reduce the need for large, annotated datasets.

2. **Enhanced Performance:** Benefits from rich feature representations learned from extensive datasets.
3. **Reduced Training Time:** Speeds up training by starting with pre-trained weights.

Selection of Pre-trained Models

Chosen Models: MobileNet, ResNet50, VGG16, InceptionV3

Criteria:

- **Diverse Architectures:** Provides a comprehensive comparison of different deep learning approaches.
- **Proven Performance:** Demonstrated efficacy across various benchmarks.
- **Varied Computational Requirements:** Allows analysis of performance versus resource consumption.

Implementation Details

Pre-trained Models & Fine-tuning:

1. **Base Configuration:** Removed top layers, standardized input to (224, 224, 3).
2. **Freezing Layers:** Preserved pre-trained features by freezing all base layers.
3. **Custom Classification Head:** Added global pooling, multiple Dense layers (128, 64, 32 neurons with ReLU), and a softmax output layer for six classes.
4. **Compilation:** Used Adam optimizer (learning rate 1e-4), categorical cross-entropy loss, and accuracy metric.

Training Strategy:

- **Data Generators:** Utilized Keras' ImageDataGenerator for augmentation and efficient loading.
- **Batch & Epochs:** Batch size of 64, trained for 30 epochs with progress monitoring.
- **Callbacks:** Applied ReduceLROnPlateau to adjust learning rates dynamically.
- **Model Saving:** Saved models with architecture-specific filenames.

Rationale for Method Selection

- **Diverse Architectures:** Ensures thorough evaluation of feature extraction and classification methods.
- **Pre-trained Efficiency:** Accelerates development and leverages existing features.
- **Preventing Overfitting:** Freezing layers reduces overfitting risks with limited data.
- **Task-specific Heads:** Custom layers enhance accuracy for mosquito detection.

Lightweighting Strategies

MobileNet:

- Quantization, pruning, and knowledge distillation to reduce size and improve speed.

ResNet50:

- Pruning, layer reduction (e.g., ResNet34), quantization, and efficient residual blocks.

VGG16:

- Network pruning, parameter sharing, knowledge distillation, and replacing large dense layers.

InceptionV3:

- Module optimization, pruning, quantization, simplifying auxiliary classifiers, and adaptive computation.

IV. Experimental Setup

Model Parameters:

Parameter	Setting	Purpose
Learning Rate Adjustment Callback	monitor='val_loss', patience=2, factor=0.5	Reduces the learning rate if validation loss does not improve, helping the model escape local optima.
Batch Size	batch_size=64	Affects training speed and stability.
Target Image Size	target_size=(224, 224)	Resizes images to match model input requirements.
Class Mode	class_mode='categorical'	Sets up multi-class classification.
Number of Epochs	epochs=30	Determines training duration. Higher values may risk overfitting.

Data Split

Train folder: Contains data for model training.

Test folder: Contains data for model evaluation.

Pred folder: Contains data for prediction or validation.

V. Results

In the study of mosquito recognition of human skin, we chose four deep learning models for experiment. These models include MobileNet, ResNet50, VGG16 and InceptionV3. The experiment result are shown in Table 1. ResNet50 performed best among all models with a validation accuracy of 94% and a training loss and validation loss of 0.15 and 0.25, respectively. This followed by MobileNet, which achieved 90% verification accuracy. VGG16 was 88 percent accurate. The final InceptionV3 was the weakest model, with only 75% accuracy. It means that ResNet50 and MobileNet generalize better on this dataset. The low accuracy of InceptionV3 may be due to the complexity of its model result, which may lead to overfitting or underfitting problem during the experiment.

Model	Training Accuracy	Validation Accuracy	Training Loss	Validation Loss
MobileNet	0.93	0.9	0.25	0.35
ResNet50	0.96	0.94	0.15	0.25
VGG16	0.91	0.88	0.3	0.45
InceptionV3	0.78	0.75	0.5	0.55

Table 1 Model Performance Result

The confusion matrices give us more details. MobileNet and VGG16 did well with the Culex categories. They had low misclassification rates. But they had trouble with the aegypti and albopictus categories. It was especially hard for them to tell the difference between aegyptie smashed and albopictus landing. ResNet50 made fewer mistake in these cases, which explains its higher validation accuracy. The confusion be because these categories look similar. This makes it hard for the models to pick the right features.

To check the system's performance, we need to look at more than just accuracy. We also need to look at the loss values. Both ResNet50 and MobileNet had stable loss trends. Their validation losses is below 0.35. This means there was no serious overfitting. On the other hand, VGG16 and InceptionV3 had higher losses. This suggest that they did not learn the features as well. InceptionV3's training and validation loss were 0.50 and 0.55. This show underfitting and poor feature learning on this dataset.

When we compare these models to current top methods (SOTA), ResNet50's performance was strong. It showed high validation accuracy and low loss values. MobileNet also did well because it have a lightweight design. This make it good for use in situations with limited resources. However, VGG16 and InceptionV3 did not perform as well. Even though they were successful in the past, their large size and complex structure may not be a good fit for this dataset.

Overall, ResNet50 perform best in our projects, followed by MobileNet. Although VGG16 and InceptionV3 are weak, this may not be problem of the model itself, it may defect of the algorithm, and our data processing of the model is not good enough. In future work, we plan to adapt architecture of the experiment model and apply advanced data enhancements to improve its performance. We can try to use test models like EfficientNet or Vision Transformer improve accuracy and versatility. To sum up, we show the performance of these four model result. ResNet50 is most effective, while MobileNet strike good balance between accuracy and efficiency. Our future research should focus on updating the architecture and lightweight optimizations to try further improve the performance of the model.

VI. Conclusions

1. Contribution of this study

This paper compares the performances of four deep learning models, including MobileNet, ResNet50, VGG16, and InceptionV3, for classification tasks and comprehensively analyzes the differences in accuracy, speed of training, and recognition, as well as resource requirements of the mentioned models. These models represent different architectures, such as the light-weight model represented by MobileNet, the residual network represented by ResNet50, the classic convolutional neural network represented by VGG16, and multi-scale convolutional neural network represented by InceptionV3. These results provide the solid basis for how to select appropriate deep learning models in practical applications, especially when decisions are made under different resource constraints and real-time requirements. In particular,

the results of this study provide an important reference value for researchers and developers who want to implement image classification on embedded devices, servers, and other platforms.

2. Each model performance

In the test set, the best performance was ResNet50, with an accuracy of 93.66%, though quite good during training, also in recognition speed, finding applications where there is a high-accuracy requirement, but the computational resources are sufficient. VGG16 possesses relatively high accuracy, 91.77%. However, because the model architecture is complicated and the parameters are too large, the training and recognition speeds are relatively worse; thus, it can be more suitable for offline processing tasks. Though having a bit lower accuracy, 88.77%, for its lightweight design, the training and recognition speed were very fast, and it is suitable for scenes where resources are constrained and real-time applications need to be performed, including on mobile terminals and embedded devices. The worst performance within this study was given by InceptionV3, which only reached an accuracy of 76.21%. Its training and recognition speed is slow; probably not suitable for this dataset or having overfitting problems.

3. Directions of Future Improvement

1) Expansion and optimization of the dataset: In the future, when generalizing the performance of categories with more misclassifications, like the aegypti and albopictus categories, the size of the dataset can be improved. At the same time, more samples with discriminative features can be added in the training set to help the model to identify the subtle differences better.

2) Data augmentation technology: More methods can be used to enhance the image data, such as color transformation, random cropping, or adding random noise, in order to improve robustness, especially for categories with fewer samples, thus helping to reduce overfitting of the model.

3) Model architecture optimization: Considering the advantages of MobileNet with ResNet, a new lightweight model or adaptive architecture may be worthwhile, further reducing the computation amount and improving accuracy. Besides, a multi-task learning mechanism could also be introduced for enhancing the discrimination ability between categories through jointly learning multiple related tasks.

4) Automatic model selection and tuning: Introduce automated model selection and hyperparameter tuning algorithms that perform parameter and layer structure optimization to select an appropriate architecture for a given task.

5) In a word, this paper compared four mainstream models in terms of performance and gave the conclusion that MobileNet is the best choice when resources are limited, while ResNet50 can be more suitable when considering accuracy. Directions of future improvement give guiding suggestions to improve the accuracy and applicability of a model.

VII. References

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